

EXTENSION OF EFFICIENT LOW DISSIPATION HIGH ORDER SCHEMES FOR 3-D CURVILINEAR MOVING GRIDS¹

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Abstract

The efficient low dissipative highly parallelizable shock-capturing schemes of essentially fourth-order or higher proposed by Yee et al. (1999a) is formulated for 3-D curvilinear moving grids in the finite-difference frame work. These schemes consist of high order compact or non-compact non-dissipative base schemes combined with adaptive nonlinear characteristic filters to minimize the use of numerical dissipation away from shock and shear regions. The amount of numerical dissipation is further minimized by applying these schemes to the entropy splitting form of the inviscid flux derivatives. The analysis is given for a thermally perfect gas. The main difficulty in the extension of high order schemes to curvilinear moving grids is the high order numerical evaluation of the geometric terms arising from the coordinate transformation. The numerical evaluation of these terms to insure freestream preservation is done in a coordinate invariant manner. This avoids spurious numerical errors, which would result from previous, noninvariant formulations, when treating axi-symmetric flow.

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I. Introduction

Most available high order high-resolution shock-capturing numerical schemes are too CPU intensive for practical 3-D complex simulations. In spite of their high-resolution capability for rapidly evolving flows and short term time integrations, these schemes often exhibit undesirable amplitude errors for long time integrations in aeroacoustics, rotorcraft, turbulence and general long wave propagation computations. High order here refers to spatial schemes that are essentially fourth-order or higher away from shock and shear regions. The delicate balance of the numerical dissipation for stability without the expense of excessive smearing of the flow physics after long time integrations poses a severe challenge for unsteady flow simulations of this type. The recently developed high order low-dissipative shock capturing schemes of Yee et al. (1999a) and their companion papers Yee et al. (1999b) and Sjogreen & Yee (2000) aim at developing methods to overcome some of the difficulties. For efficiency, Yee et al. (1999a) proposed simple highly parallelizable spatial schemes that consist of a base scheme and nonlinear filters. The base scheme consists of narrow grid stencil high order compact or non-compact centered non-dissipative classical spatial differencings. The filters consist of a product of the dissipative portion of low order total variation diminishing (TVD), essentially non-oscillatory (ENO) or weighted ENO (WENO) schemes and an artificial compression method (ACM) sensor. The role of the ACM sensor is to reduce the amount of numerical dissipation away from shock and shear regions. As an alternative to the ACM sensor, Sjogreen & Yee (2000) utilized non-orthogonal wavelet basis functions as multi-resolution sensors to dynamically determine the amount of nonlinear numerical dissipation to be added at each grid point. The resulting wavelet sensors are readily available as more desirable grid adaptation indicators (Gerritsen & Olsson 1996) than the commonly used grid adaptation indicators.

In contrast to hybrid schemes that switch between spectral or spectral-like non-shock-capturing schemes and high order ENO schemes, the high order non-dissipative base scheme is always activated. The final grid stencil of these schemes is five points in each spatial direction if second-order TVD schemes are used as filters, and seven points if second-order ENO schemes are used as filters for a fourth-order base scheme. Studies showed that higher accuracy was achieved with less CPU time and fewer grid points when compared with that of standard high order TVD, positive, ENO or WENO schemes. These schemes are able to accurately simulate a wide range of flow conditions, including long time integrations of wave propagation, computational aeroacoustics, combustion and direct numerical simulation (DNS) of 3-D compressible turbulence. See Yee et al. (1999a,b), Yee et al. (2000), Sandham & Yee (2000), Sjogreen & Yee (2000, 2001), Müller & Yee (2001) and Polifke et al. (2001). Table 1.1 shows the flow chart of the schemes.

In Yee et al. (1999b), these schemes were applied to the entropy splitting form of the inviscid flux derivatives (e.g., F_x ; see bottom of Table 1.1.) Studies were conducted to determine to what extent the entropy splitting form of the flux derivative can help in minimizing numerical dissipation, or equivalently, in improving nonlinear stability. They view entropy splitting as a conditioned form of the original conservation laws. Overall, the numerical results of Yee et al. (1999b), Sandham & Yee (2000), Sjogreen & Yee (2000) and Polifke et al. (2001) indicate a positive benefit from the entropy splitting. The splitting can stabilize spurious noise generated by the non-dissipative or low dissipative spatial discretizations which is a major cause of nonlinear instability. Their study also indicates that entropy splitting **alone** can improve

nonlinear stability even when one employs numerical boundary conditions that do not satisfy the Strand (1994) condition. This stability property of the entropy splitting is valuable not just for the class of schemes in question, but can also be applied to other schemes commonly used in practice. This emphasizes the fact that one should always try to apply numerical schemes to a more conditioned form of the governing equations.

In Sandham & Yee (2000), entropy splitting with a variant of Strand’s boundary difference operators that were developed by Carpenter et al. (1998) was used in conjunction with the high order non-dissipative central base scheme for a DNS of 3-D shock-free compressible turbulent channel flow. Very accurate fully developed turbulent statistics were obtained using coarse to moderate grid sizes and without using filters. Their results compared well with the best spectral method of incompressible Navier-Stokes simulations. Modern high-resolution numerical dissipation has been the major factor in improving nonlinear instabilities for short or moderate time integrations (unsteady). Most often, added numerical dissipation is necessary for longer time integration at the expense of excess smearing of the flow physics without resorting to finer grids and extremely small time steps. The use of the entropy splitting form of the flux derivative was shown to be capable of minimizing the use of numerical dissipation.

In Yee et al. (1999b), the extendibility of the entropy splitting concept to other physical equations of state and evolutionary equation sets was examined. Their study shows that the entropy splitting can be formally extended to a thermally perfect gas, with the internal energy being an arbitrary function of temperature. Although extension of entropy splitting to nonequilibrium flows is not practically feasible, and is not possible for equilibrium real gas and artificial compressibility methods of solving the incompressible Navier-Stokes equations, and magnetohydrodynamic (MHD) flows, the high order numerical schemes in question are applicable to these types of flows. Note that since the Maxwell equations are a linear system of hyperbolic equations that can be easily symmetrized, Strand’s numerical boundary operators are still valid and the numerical schemes in question are also applicable. Entropy splitting is not needed for the Maxwell equations. For nonequilibrium flows, if one solves the species and flow equations separately in a loosely coupled manner, then the flow equations effectively satisfy a locally thermally perfect gas law and a “local” form of entropy splitting is applicable.

In order to apply this scheme in practice, it must be formulated in arbitrary 3-D curvilinear coordinates. A finite-volume formulation is generally preferred over a finite-difference formulation, especially for schemes that are third-order or lower. For fourth-order or higher schemes, finite-volume formulations are very complex. For efficiency, we therefore only consider finite-difference formulations. An important issue then is the treatment of the geometric terms arising from the coordinate transformation. These are the components of the surface area vectors (or metrics) appearing in the transformed fluxes and the volume (or Jacobian) in the definition of the transformed conservative variables. Analytically, these geometric quantities satisfy certain conservation laws. They are the surface area conservation law (or metric identity), and for time-varying grids an additional volume conservation law (sometimes referred to as the geometric conservation law). It would be desirable for these laws to be satisfied numerically. This would result in regions of uniform flow being preserved exactly (within round-off errors). In regions of non-uniform flow, it would hopefully lead to greater accuracy by eliminating errors due to the grid, particularly if it is highly distorted.

In Vinokur (1989), it was shown how these two laws can be satisfied numerically by making

use of finite-volume concepts. For the surface area vector components, the discretization was found to be identical to the averaging procedure proposed by Pulliam & Steger (1978) in the second-order finite-difference frame work. Unfortunately, this use of finite-volume concepts is only valid for second-order accurate central differencing. They cannot be extended to high order compact or non-compact differencings. An alternative method of discretizing the surface area vector components by first rewriting them in an equivalent "conservative" form was proposed by Thomas & Lombard (1979). For second-order accurate central differencing, the surface area conservation law was then numerically satisfied. Gaitonde & Visbal (1999) found by numerical experiments on two different curvilinear grids that high order compact and non-compact differencing applied to the Thomas and Lombard form also satisfied that conservation law numerically. An undesirable feature of the expression used is that it is not coordinate invariant. The present coordinate invariant form has been independently proposed by the authors in June 2000 and by Thomas & Neier (1990) (P.D. Thomas, private communication, November 2000).

Objectives: In this paper we formulate the Yee et al. (1999a) scheme with entropy splitting (Yee et al. 1999b) in 3-D curvilinear coordinates for a thermally perfect gas in a finite-difference approach. The surface area conservation law is satisfied numerically by discretizing a coordinate invariant version of the Thomas and Lombard formulas. The importance of doing this is that it enables us to extend the method to axi-symmetric flow, with an appropriate treatment of the resulting source term. This eliminates spurious numerical errors due to the original Thomas and Lombard expressions. We also provide a rigorous analytic proof that the numerical mixed partial derivatives commute for any grid, using any high order compact or non-compact differencing, with arbitrary numerical boundary conditions. This is necessary to insure that the surface area conservation law is satisfied numerically.

Outline: In Section II we review the method of entropy splitting as applied to the Euler equations for a thermally perfect gas. The equations are formulated for arbitrary 3-D curvilinear coordinates, and specialized to axi-symmetric flow. The section also includes a brief discussion of the Navier-Stokes equations. In Section III we describe the numerical methods to treat the spatial terms. Particular attention is paid to the treatment of the geometric terms. A section on Roe's approximate Riemann solver, which is part of the numerical method for the filters, includes a discussion of some of the issues involved in defining the eigenvectors of the flux Jacobian matrix for generalized 3-D coordinates, as well as the expressions for the Roe-averaged state for a thermally perfect gas. The corresponding forms for nonequilibrium flows are presented in Appendix B. The proof that numerical mixed partial derivatives commute for the high order base scheme in generalized coordinates is presented in Appendix A.

II. Formulation of Equations

2.1. Canonical Splitting of Conservation Laws

A system of scalar conservation laws can be written as

$$Q_t + \nabla \cdot \mathbf{F} = 0, \quad (2.1.1)$$

where Q and $\mathbf{F}(Q)$ are algebraic *vectors*, but the components of $\mathbf{F}$ are physical vectors. Letter subscripts indicate partial differentiation. In order to obtain a nonlinearly stable method

of solving initial boundary value problems (IBVPs) for the nonlinear system of hyperbolic conservation laws (2.1.1), we transform the governing equations so that the resulting PDEs are nonlinearly stable, including the effect of physical boundary conditions (Olsson & Oliger (1994)). We introduce the new *vector* $W(Q)$ such that $\mathbf{F}_W$ is symmetric and Q_W is symmetric and positive definite. Here the matrix $\mathbf{F}_W$ also has components that are physical vectors. Furthermore, W is chosen such that both $\mathbf{F}$ and Q are homogeneous functions of W of degree β , i.e., there is a constant β such that for all σ

$$Q(\sigma W) = \sigma^\beta Q(W), \quad (2.1.2a)$$

$$\mathbf{F}(\sigma W) = \sigma^\beta \mathbf{F}(W). \quad (2.1.2b)$$

By taking the partial derivative of Eqs. (2.1.2) with respect to σ and then setting $\sigma = 1$, we obtain Euler's theorem, namely

$$Q_W W = \beta Q, \quad (2.1.3a)$$

$$\mathbf{F}_W W = \beta \mathbf{F}. \quad (2.1.3b)$$

In order to obtain an energy estimate for stability of the IBVP we introduce a canonical splitting and write $\nabla \cdot \mathbf{F}$ as

$$\nabla \cdot \mathbf{F} = \frac{\beta}{\beta+1} \nabla \cdot \mathbf{F} + \frac{1}{\beta+1} \mathbf{F}_W \cdot \nabla W, \quad \beta \neq -1, \quad (2.1.4)$$

where ∇W is a *vector* whose components are the gradients of the components of W . The “ $\cdot$ ” in the second term in (2.1.4) indicates that when taking the inner product of a row of $\mathbf{F}_W$ and the *vector* ∇W , the vector dot product is applied when multiplying components. Note that $\nabla \cdot \mathbf{F}$ has been split into a conservative part and a non-conservative part. If we take the inner product of W and $\nabla \cdot \mathbf{F}$, and use the homogeneity property of $\mathbf{F}$ and the symmetry property of $\mathbf{F}_W$, we obtain

$$W^T \nabla \cdot \mathbf{F} = \frac{1}{\beta+1} [W^T \beta \nabla \cdot \mathbf{F} + W^T \mathbf{F}_W \cdot \nabla W] \quad (2.1.5a)$$

$$= \frac{1}{\beta+1} [W^T \nabla \cdot (\mathbf{F}_W W) + \nabla W^T \cdot \mathbf{F}_W W] \quad (2.1.5b)$$

$$= \frac{1}{\beta+1} \nabla \cdot (W^T \mathbf{F}_W W). \quad (2.1.5c)$$

Applying the canonical splitting to Q_t , we obtain in a similar manner

$$W^T Q_t = \frac{1}{\beta+1} (W^T Q_W W)_t. \quad (2.1.6)$$

Taking the inner product of W with (2.1.1), substituting (2.1.5c) and (2.1.6), integrating over a fixed domain with boundary surface element $\mathbf{n} dS$, where $\mathbf{n}$ is the outward normal, and applying the divergence theorem, results in

$$\frac{d}{dt} \int W^T Q_W W dV = - \oint W^T \mathbf{n} \cdot \mathbf{F}_W W dS. \quad (2.1.7)$$

Here $\mathbf{n} \cdot \mathbf{F}_W$ is the matrix whose components are the outward normal vector components of $\mathbf{F}_W$. From (2.1.7) one can rigorously establish a bound on the rate of growth of the energy norm in terms of the absolute eigenvalues corresponding to the incoming characteristic variables at the boundary of the domain (Olsson & Olinger 1994, Gerritsen & Olsson 1996). In doing so, one must make use of the positive definite property of Q_W .

2.2. Entropy Splitting of the Euler Equations

In this section we consider the conservation-law form of the 3-D compressible Euler equations. For maximum generality, the analysis is presented for arbitrary three-dimensional, time varying grids. The canonical splitting discussed in Section 2.1 is based on a convex entropy function that satisfies an entropy condition. The splitting is therefore referred to as entropy splitting. While this splitting was originally derived for a perfect gas (Gerritsen & Olsson 1996), it can be extended to flow of a gas that is only thermally perfect, with the internal energy being an arbitrary function of temperature (Yee et al. 1999b). This law is valid for a dilute gas consisting of a single chemical species, and is also a very good approximation for air below the temperature when oxygen starts to dissociate (approximately 2000° K).

Let $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ be the unit vectors along the axes of a Cartesian coordinate system. Then the position vector $\mathbf{r}$ and flow velocity $\mathbf{u}$ are expressed as

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (2.2.1)$$

and

$$\mathbf{u} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}. \quad (2.2.2)$$

The grid point velocity in a time varying grid is designated by $\mathbf{v}$. Let $\mathbf{u}' = \mathbf{u} - \mathbf{v}$ be the velocity relative to the moving grid. The *vectors* Q and $\mathbf{F}$ for the Euler equations then take the form

$$Q = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ e \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho \mathbf{u}' \\ \rho u \mathbf{u}' + p\mathbf{i} \\ \rho v \mathbf{u}' + p\mathbf{j} \\ \rho w \mathbf{u}' + p\mathbf{k} \\ (e + p)\mathbf{u}' + p\mathbf{v} \end{bmatrix}, \quad (2.2.3)$$

where ρ is the density, p is the pressure, and $e = \rho(\epsilon + q^2/2)$ is the total energy per unit volume. ϵ is the specific internal energy and $q^2 = u^2 + v^2 + w^2$ is the square of the speed. Note that $\mathbf{F}$ is now defined relative to the moving grid.

The equation of state for a thermally perfect gas is

$$p = \rho \hat{T}, \quad (2.2.4)$$

where $\hat{T} = RT$ is a normalized temperature and R is the gas constant. $\hat{T}$ has the same dimension as ϵ . From the first and second laws of thermodynamics we can show that $\epsilon = \epsilon(\hat{T})$ only. All real gases satisfy the conditions $\dot{\epsilon} > 0$ and $\ddot{\epsilon} > 0$, where $\dot{\epsilon} = d\epsilon/d\hat{T}$. The temperature $\hat{T}(Q)$ is obtained by solving the equation

$$\epsilon(\hat{T}) = \frac{e}{\rho} - \frac{1}{2} \frac{[(\rho u)^2 + (\rho v)^2 + (\rho w)^2]}{\rho^2}. \quad (2.2.5)$$

Equation (2.2.5) has a unique solution since $\dot{\epsilon} > 0$.

From the laws of thermodynamics we can relate the dimensionless entropy $\hat{\mathcal{S}} = \mathcal{S}/R$ to ρ and $\hat{T}$ by

$$\rho f = e^{-\hat{\mathcal{S}}}, \quad (2.2.6)$$

where

$$f(\hat{T}) = \exp\left(-\int \frac{\dot{\epsilon}}{\hat{T}} d\hat{T}\right). \quad (2.2.7)$$

The arbitrary constant in the integral of (2.2.7) can be absorbed in the definition of $\hat{\mathcal{S}}$.

Following Harten (1983a), we obtain the *vector* $W(Q)$ from

$$W = \frac{\partial \Gamma}{\partial Q}, \quad (2.2.8)$$

where the convex function $\Gamma(Q)$ is given by

$$\Gamma = \rho \psi(\hat{\mathcal{S}}). \quad (2.2.9)$$

The components of W are sometimes referred to as “entropy variables”, while Γ is referred to as an “entropy function”. We show in Yee et al. (1999b) that in order to satisfy the homogeneity and positive definite conditions, $\psi(\hat{\mathcal{S}})$ is given by

$$\psi = \beta e^{-\hat{\mathcal{S}}/\beta}, \quad (2.2.10)$$

where β is a constant. This then gives

$$\dot{\psi} = -e^{-\hat{\mathcal{S}}/\beta} = -(\rho f)^{1/\beta}, \quad (2.2.11)$$

where we have used (2.2.6) to obtain the second expression. We note from (2.1.4) that the definition of W allows an arbitrary multiplicative constant. Correspondingly, there is an arbitrary multiplicative constant in the definition of ψ . We have chosen this constant to obtain the simplest form for (2.2.11). The transformed variable W can now be written as

$$W = \frac{\dot{\psi}}{p} [e - 2\tilde{\epsilon} - p(1 + \beta) \quad -\rho u \quad -\rho v \quad -\rho w \quad \rho]^T, \quad (2.2.12)$$

where $\tilde{\epsilon} = \rho \epsilon$ is the internal energy per unit volume. With the aid of (2.2.4), we can combine the components of W to show that $\hat{T}$ and $\epsilon(\hat{T})$ are homogeneous functions of W of degree 0. It then follows from (2.2.3), (2.2.4), (2.2.11) and (2.2.12) that the *vectors* Q and $\mathbf{F}$ are both homogeneous functions of W of degree β .

While we cannot obtain an explicit formula for $Q(W)$, we can derive an explicit expression for the symmetric matrix Q_W as a function of Q . The upper triangular part can be written as

$$Q_W = \frac{1}{\dot{\psi}} \begin{bmatrix} a\rho & a\rho u & a\rho v & a\rho w & ae + bp \\ & a\rho u^2 - p & a\rho uv & a\rho uw & u[ae + (b-1)p] \\ & & a\rho v^2 - p & a\rho vw & v[ae + (b-1)p] \\ & & & a\rho w^2 - p & w[ae + (b-1)p] \\ & & & & \frac{ae^2}{\rho} + p\left(\frac{2be}{\rho} - q^2\right) + \frac{bp^2}{\rho}(1 + \beta) \end{bmatrix}, \quad (2.2.13)$$

where

$$a(\hat{T}, \beta) = \frac{1}{\dot{\epsilon} + 1 + \beta} - 1 \quad (2.2.14)$$

and

$$b(\hat{T}, \beta) = (1 + \beta)a + \beta = \frac{-\dot{\epsilon}}{\dot{\epsilon} + 1 + \beta}. \quad (2.2.15)$$

We prove in Yee et al. (1999b) that the positive definite condition on Q_W requires that $\dot{\psi} < 0$, a condition already satisfied by (2.2.11), and that

$$\frac{1}{\beta} > \frac{-1}{1 + \dot{\epsilon}}. \quad (2.2.16)$$

Condition (2.2.16) is satisfied if $\beta > 0$ or $\beta < -(1 + \dot{\epsilon})$. Since $\ddot{\epsilon} > 0$, the maximum value of $\dot{\epsilon}$ occurs at $\hat{T}_{max}$. Therefore, for $\beta < 0$,

$$\beta < -[1 + \dot{\epsilon}(\hat{T}_{max})]. \quad (2.2.17)$$

A sufficiency condition, independent of the flow problem, is obtained by replacing $\dot{\epsilon}(\hat{T}_{max})$ by $\dot{\epsilon}(\infty)$.

Specialization for a Perfect Gas: For a perfect gas, with a ratio of specific heats γ , the caloric equation of state becomes

$$\epsilon = \frac{\hat{T}}{(\gamma - 1)} \quad \text{and} \quad \dot{\epsilon} = \frac{1}{\gamma - 1}. \quad (2.2.18)$$

It follows that

$$\rho f = (p\rho^{-\gamma})^{\frac{-1}{(\gamma-1)}}, \quad (2.2.19)$$

$$\dot{\psi} = -(p\rho^{-\gamma})^{\frac{-1}{(\gamma-1)\beta}}, \quad (2.2.20)$$

$$a(\gamma, \beta) = \frac{\gamma - 1}{\gamma + (\gamma - 1)\beta} - 1, \quad (2.2.21)$$

and

$$b(\gamma, \beta) = \frac{-1}{\gamma + (\gamma - 1)\beta}. \quad (2.2.22)$$

The positive definite condition on β then becomes $\beta > 0$ or $\beta < \frac{-\gamma}{\gamma-1}$. See Yee et al. (1999b) and Sandham & Yee (2000) for a study on the beneficial ranges of β for a variety of flows.

2.3. Formulation in Generalized Curvilinear Coordinates

The equations presented so far can, in principle, be implemented by any numerical method, using any type of grid. Since our interest lies in efficient, high order accurate solutions, we will limit ourselves to finite difference formulation on a structured grid. In this section we therefore consider the formulation of the equations in generalized curvilinear coordinates. We first present the equations for a three-dimensional flow. They will then be specialized to the

important case of axi-symmetric flow. The section concludes with a brief discussion of the Navier-Stokes equations.

2.3.1. Three-Dimensional Flow

An arbitrary, time-dependent transformation from curvilinear coordinates to physical space is written as

$$\mathbf{r} = \mathbf{r}(\xi, \eta, \zeta, \tau), \quad (2.3.1a)$$

$$t = \tau. \quad (2.3.1b)$$

For the computational cell $d\xi, d\eta$ and $d\zeta$, the normalized surface area vectors in the ξ, η, ζ directions are given by

$$\mathbf{S}^\xi = \mathbf{r}_\eta \times \mathbf{r}_\zeta, \quad \mathbf{S}^\eta = \mathbf{r}_\zeta \times \mathbf{r}_\xi, \quad \mathbf{S}^\zeta = \mathbf{r}_\xi \times \mathbf{r}_\eta. \quad (2.3.2)$$

The normalized cell volume is given by

$$\mathcal{V} = \mathbf{r}_\xi \cdot \mathbf{r}_\eta \times \mathbf{r}_\zeta \quad (2.3.3)$$

and the grid point velocity is given by

$$\mathbf{v} = \mathbf{r}_\tau. \quad (2.3.4)$$

Applying transformation (2.3.1) to the moving grid version of (2.1.1) we obtain

$$\hat{Q}_\tau + \hat{E}_\xi + \hat{F}_\eta + \hat{G}_\zeta = 0, \quad (2.3.5a)$$

where

$$\hat{Q} = \mathcal{V}Q, \quad \hat{E} = \mathbf{S}^\xi \cdot \mathbf{F}, \quad \hat{F} = \mathbf{S}^\eta \cdot \mathbf{F}, \quad \hat{G} = \mathbf{S}^\zeta \cdot \mathbf{F}. \quad (2.3.5b)$$

In what follows we will use numerical subscripts to indicate Cartesian components. Thus

$$\mathbf{S}^\xi = S_1^\xi \mathbf{i} + S_2^\xi \mathbf{j} + S_3^\xi \mathbf{k}, \quad (2.3.6)$$

with similar definitions for $\mathbf{S}^\eta$, $\mathbf{S}^\zeta$, and $\mathbf{v}$. Let

$$\hat{v}^\xi = \mathbf{S}^\xi \cdot \mathbf{v} = S_1^\xi v_1 + S_2^\xi v_2 + S_3^\xi v_3, \quad (2.3.7)$$

with similar definitions for $\hat{v}^\eta$ and $\hat{v}^\zeta$. For the Euler equations, the transformed flux $\hat{E}$ is given by

$$\hat{E} = \begin{bmatrix} \rho \hat{U} \\ \rho u \hat{U} + S_1^\xi p \\ \rho v \hat{U} + S_2^\xi p \\ \rho w \hat{U} + S_3^\xi p \\ (e + p) \hat{U} + \hat{v}^\xi p \end{bmatrix}, \quad (2.3.8)$$

where

$$\widehat{U} = \mathbf{S}^\xi \cdot \mathbf{u}' = S_1^\xi u + S_2^\xi v + S_3^\xi w - \widehat{v}^\xi. \quad (2.3.9)$$

The transformed flux derivative $\widehat{E}_\xi$ is now split as

$$\widehat{E}_\xi = \frac{\beta}{\beta+1} \widehat{E}_\xi + \frac{1}{\beta+1} E_W^* W_\xi, \quad (2.3.10)$$

where $E_W^* = \mathbf{S}^\xi \cdot \mathbf{F}_W$ is given by

$$E_W^* = \frac{1}{\dot{\psi}} \begin{bmatrix} a\rho\widehat{U} & a\rho u\widehat{U} - S_1^\xi p & a\rho v\widehat{U} - S_2^\xi p & a\rho w\widehat{U} - S_3^\xi p & e_{15} \\ (a\rho u^2 - p)\widehat{U} - 2uS_1^\xi p & a\rho uv\widehat{U} - p(S_2^\xi u + S_1^\xi v) & e_{24} & e_{25} \\ (a\rho v^2 - p)\widehat{U} - 2vS_2^\xi p & e_{34} & e_{35} \\ e_{44} & e_{45} \\ e_{55} \end{bmatrix}, \quad (2.3.11)$$

and

$$e_{15} = [ae + (b-1)p]\widehat{U} - \widehat{v}^\xi p, \quad (2.3.12a)$$

$$e_{24} = a\rho uw\widehat{U} - p(S_3^\xi u + S_1^\xi w), \quad (2.3.12b)$$

$$e_{25} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}u - \frac{p}{\rho}(e+p)S_1^\xi, \quad (2.3.12c)$$

$$e_{34} = a\rho vw\widehat{U} - p(S_3^\xi v + S_2^\xi w), \quad (2.3.12d)$$

$$e_{35} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}v - \frac{p}{\rho}(e+p)S_2^\xi, \quad (2.3.12e)$$

$$e_{44} = (a\rho w^2 - p)\widehat{U} - 2wS_3^\xi p, \quad (2.3.12f)$$

$$e_{45} = \{[ae + (b-2)p]\widehat{U} - \widehat{v}^\xi p\}w - \frac{p}{\rho}(e+p)S_3^\xi, \quad (2.3.12g)$$

$$e_{55} = \left[\frac{ae^2}{\rho} + p\{2(b-1)\frac{e}{\rho} - q^2\} + \frac{p^2}{\rho}\{b(1+\beta) - 2\}\right]\widehat{U} - 2\widehat{v}^\xi \frac{p}{\rho}(e+p). \quad (2.3.12h)$$

The analogous expressions for $\widehat{F}$ and $\widehat{F}_\eta$ are obtained from (2.3.8) through (2.3.12) by replacing $\widehat{U}$ with $\widehat{V}$ and ξ with η throughout. Similarly, the expressions for $\widehat{G}$ and $\widehat{G}_\zeta$ are obtained by replacing $\widehat{U}$ with $\widehat{W}$ and ξ with ζ throughout. Here $\widehat{V}$ and $\widehat{W}$ are (2.3.9) with $\mathbf{S}^\xi$ replaced by $\mathbf{S}^\eta$ and $\mathbf{S}^\zeta$ respectively, and $\widehat{W}$ has no relationship to the entropy splitting vector W in (2.2.12).

Normally, we need to compute $\widehat{Q}_W$ for the split form of $\widehat{Q}_\tau = \frac{\beta}{\beta+1}\widehat{Q}_\tau + \frac{1}{\beta+1}\widehat{Q}_W W_\tau$. However, we only consider a semi-discrete approach of applying temporal discretizations. Aside from using the split form of the inviscid flux derivatives $\widehat{E}_\xi$, $\widehat{F}_\eta$ and $\widehat{G}_\zeta$, we do not have to use the split form of $\widehat{Q}_\tau$ for implementation. Thus the final form of the semi-discrete entropy splitting approach still can be expressed in terms of conservative and primitive variables, making possible easy and efficient implementation in existing computer codes.

From definitions (2.3.2) we can derive the
Surface Area Conservation Law

$$(\mathbf{S}^\xi)_\xi + (\mathbf{S}^\eta)_\eta + (\mathbf{S}^\zeta)_\zeta = 0, \quad (2.3.13)$$

which is valid for each of the Cartesian components. For time-varying grids, by combining (2.3.13) with (2.3.5), and assuming a uniform flow, we derive the

Volume Conservation Law

$$\mathcal{V}_\tau = (\hat{v}^\xi)_\xi + (\hat{v}^\eta)_\eta + (\hat{v}^\zeta)_\zeta. \quad (2.3.14)$$

Note that we have not written relations (2.3.2) in their component forms. We will show in Section 3.3.1 that in order to satisfy (2.3.13) numerically, these relations must be modified. Finally, we relate our notation to the more familiar one introduced by Steger (1977). These are

$$\mathcal{V} = J^{-1}, \quad \hat{U} = J^{-1}U, \quad S_1^\xi = J^{-1}\xi_x, \quad \hat{v}^\xi = -J^{-1}\xi_t, \quad (2.3.15)$$

with analogous relations for the other quantities, where Steger defines J as the Jacobian of the transformation and U is his contravariant velocity in the ξ -direction.

2.3.2. Axi-Symmetric Flow

In order to obtain the equations for axi-symmetric flow, we first introduce cylindrical coordinates x, r, ζ , where the x -axis is the polar axis and ζ is the polar angle. Introducing the curvilinear coordinates ξ, η in the x - r plane, we have the transformation equations

$$x = x(\xi, \eta, \tau) \quad \text{and} \quad r = r(\xi, \eta, \tau), \quad (2.3.16)$$

and

$$y = r \cos \zeta \quad \text{and} \quad z = r \sin \zeta. \quad (2.3.17)$$

The surface area vectors can then be obtained from (2.3.2). In particular, the components of $\mathbf{S}^\zeta$ become

$$S_1^\zeta = 0, \quad S_2^\zeta = -S^\zeta \sin \zeta, \quad S_3^\zeta = S^\zeta \cos \zeta, \quad (2.3.18)$$

where

$$S^\zeta \equiv x_\xi r_\eta - r_\xi x_\eta. \quad (2.3.19)$$

The cell volume is given by (2.3.3) as

$$\mathcal{V} = r S^\zeta. \quad (2.3.20)$$

For axi-symmetric flow, the ζ -components of $\mathbf{u}$ and $\mathbf{v}$, as well as the ζ derivatives of physical quantities, are all equal to 0. It follows that $\hat{v}^\zeta = \widehat{W} = 0$. The transformed flux $\hat{G}$ then becomes

$$\hat{G} = S^\zeta p [0 \quad 0 \quad -\sin \zeta \quad \cos \zeta \quad 0]^T, \quad (2.3.21)$$

while its ζ derivative is

$$\hat{G}_\zeta = -S^\zeta p [0 \quad 0 \quad \cos \zeta \quad \sin \zeta \quad 0]^T. \quad (2.3.22)$$

The axi-symmetric equations are obtained by setting $\zeta = 0$. Then the ζ -component of $\mathbf{u}$ becomes w , which is set equal to 0. It follows from (2.3.17) that $z = 0$ and r can be replaced by y in (2.3.16), (2.3.19), and (2.3.20). $\hat{G}_\zeta$ in (2.3.4) then becomes the source term

$$\hat{G}_\zeta = -S^\zeta p [0 \quad 0 \quad 1 \quad 0 \quad 0]^T. \quad (2.3.23)$$

Finally, the fourth components of all *vectors*, and fourth rows and columns of all matrices can be eliminated. Note that all the relations for the surface area components must be modified before numerical implementation, as will be indicated in Section 3.3.2. We should also mention that certain limiting forms of some relations must be used for points on the axis of symmetry.

2.3.3. Navier-Stokes Equations

So far we have treated only the Euler equations, since the canonical splitting is only applied to inviscid terms. For completeness, we make a few remarks about the Navier-Stokes equations, where the additional viscous terms are treated in a standard manner, which are well documented. The system of conservation laws is now written as

$$Q_t + \nabla \cdot (\mathbf{F} - \mathbf{F}_v) = 0. \quad (2.3.24)$$

Relative to a moving grid, $\mathbf{F}$ is given by (2.2.3), and the viscous flux *vector* $\mathbf{F}_v$ is given by

$$\mathbf{F}_v = \begin{bmatrix} 0 \\ \mathbf{i} \cdot \boldsymbol{\tau} \\ \mathbf{j} \cdot \boldsymbol{\tau} \\ \mathbf{k} \cdot \boldsymbol{\tau} \\ -q + \boldsymbol{\tau} \cdot \mathbf{u} \end{bmatrix}. \quad (2.3.25)$$

The assumption of a Newtonian fluid yields the relations

$$\boldsymbol{\tau} = \lambda(\nabla \cdot \mathbf{u})\mathbf{I} + \mu[\nabla \mathbf{u} + (\nabla \mathbf{u})^T] \quad (2.3.26)$$

and

$$q = -k \nabla T. \quad (2.3.27)$$

Here μ and λ are the first and second viscosity coefficients, and k is the thermal conductivity. $\mathbf{I}$ is the identity tensor. In order to express (2.3.25) in generalized coordinates we make use of the relation

$$\nabla = \frac{1}{\mathcal{V}} [\mathbf{S}^\xi \frac{\partial}{\partial \xi} + \mathbf{S}^\eta \frac{\partial}{\partial \eta} + \mathbf{S}^\zeta \frac{\partial}{\partial \zeta}]. \quad (2.3.28)$$

The final form of the equations in curvilinear coordinates and the corresponding high order base scheme will not be given here. They may be found in standard references. Here we will only mention the modification of the source term for axi-symmetric flow. The coefficient in (2.3.23) must be modified to account for the normal viscous stress acting in the azimuthal direction. Thus the pressure p in (2.3.23) must be replaced by

$$p - \lambda \nabla \cdot \mathbf{u} - \frac{2\mu v}{y}. \quad (2.3.29)$$

III. Numerical Methods

The spatial discretizations proposed in Yee et al. (1999a) consist of two parts, namely, a base scheme and a filter. When filters are not used, the scheme consists of only the base scheme, which is discussed in Section 3.1. The base scheme involves compact or non-compact high order central difference approximations, which are used to evaluate the flux derivatives $\widehat{E}_\xi, \widehat{F}_\eta, \widehat{G}_\zeta$, the surface area components that they contain, and the derivatives W_ξ, W_η and W_ζ . The form of the filter, which is applied to inviscid terms, is discussed in Section 3.2. The special treatment of the geometric terms necessary to satisfy the surface area conservation law numerically for the base scheme and the filter is described in Sections 3.3 and 3.4. Note that if the filters are of the same spatial order of accuracy as the base scheme, their surface area component discretizations are the same as for the base scheme. If the filters are of lower order of accuracy than the base scheme, for consistency, the corresponding order of discretization for the surface area components need only match the order of the filter.

3.1. Spatial Base Schemes

Let $\varphi_{i,j,k}$ denote the discrete approximation of a function φ at $(i\Delta\xi, j\Delta\eta, k\Delta\zeta)$, where $\Delta\xi, \Delta\eta$ and $\Delta\zeta$ are the grid spacing in the ξ, η, ζ -directions, and (i, j, k) are the corresponding spatial indices. The φ can be a function of $\widehat{Q}$ and/or (ξ, η, ζ, τ) . Possible high order central difference approximations for φ_ξ at $(i\Delta\xi, j\Delta\eta, k\Delta\zeta)$ (with analogous expressions for φ_η and φ_ζ) can be of the following two types:

Central Differencings: (fourth and sixth-order)

$$\varphi_\xi \approx \frac{-1}{12\Delta\xi} \left(\varphi_{i+2,j,k} - 8\varphi_{i+1,j,k} + 8\varphi_{i-1,j,k} - \varphi_{i-2,j,k} \right), \quad (3.1.1)$$

$$\varphi_\xi \approx \frac{1}{60\Delta\xi} \left(\varphi_{i+3,j,k} - 9\varphi_{i+2,j,k} + 45\varphi_{i+1,j,k} - 45\varphi_{i-1,j,k} + 9\varphi_{i-2,j,k} - \varphi_{i-3,j,k} \right). \quad (3.1.2)$$

Compact Central Differencings: (fourth and sixth-order)

$$\varphi_\xi \approx \frac{1}{\Delta\xi} \left(A_\xi^{-1} B_\xi \varphi \right)_{i,j,k}, \quad (3.1.3a)$$

where for a fourth-order approximation

$$(A_\xi \varphi)_{i,j,k} = \frac{1}{6} \left(\varphi_{i+1,j,k} + 4\varphi_{i,j,k} + \varphi_{i-1,j,k} \right), \quad (3.1.3b)$$

$$(B_\xi \varphi)_{i,j,k} = \frac{1}{2} \left(\varphi_{i+1,j,k} - \varphi_{i-1,j,k} \right), \quad (3.1.3c)$$

and for a sixth-order approximation

$$(A_\xi \varphi)_{i,j,k} = \frac{1}{5} \left(\varphi_{i+1,j,k} + 3\varphi_{i,j,k} + \varphi_{i-1,j,k} \right), \quad (3.1.3d)$$

$$(B_\xi \varphi)_{i,j,k} = \frac{1}{60} \left(\varphi_{i+2,j,k} + 28\varphi_{i+1,j,k} - 28\varphi_{i-1,j,k} - \varphi_{i-2,j,k} \right). \quad (3.1.3e)$$

These central difference approximations are used in applying non-dissipative high order base schemes for terms such as $\hat{E}_\xi$. For the Navier-Stokes equations, the approximations are first used to evaluate the viscous stress tensor τ and heat-flux vector q , and used again to obtain the derivatives of the viscous flux $\mathbf{F}_v$. The surface area vector components appearing in all these terms are also evaluated by using these difference approximations, but Eqs. 2.3.2 must first be modified in order to satisfy (2.3.13) exactly. This will be discussed in Section 3.3. For direct numerical simulation (DNS) of compressible turbulent computations, Sandham and Yee use a Laplacian formulation of the viscous and heat conduction terms to ensure that any tendency to odd-even decoupling associated with central schemes can be countered by the fluid viscosity. They also use a special formulation of the continuity equation based on similar arguments.

An alternate method of treating the viscous stress tensor and heat flux in the Navier-Stokes equations is to evaluate them at the half grid points. Using an appropriate high order central difference formula, one would obtain stencils for the viscous fluxes that are no wider than those for the Euler fluxes. This would require high order interpolation to evaluate the surface area vector components and transport coefficients at the half grid points. The shifted surface area vector components would not satisfy (2.3.13), but this would not affect the preservation of a uniform flow, since the viscous terms give a zero contribution. This alternate method is currently being investigated.

3.2. Filters

In this section we first review the procedure for applying the characteristic filter to multistage Runge-Kutta type and linear multistep method (LMM) types of time discretizations (Yee et al. 1999a). Examples of explicit LMMs are forward Euler and Adams-Bashforth. Examples of implicit LMMs are backward Euler, trapezoidal rule, and three-point backward differentiation. The one-leg formulation of the LMMs of Dahlquist (1979) is also applicable. We then discuss forms of the characteristic filter.

3.2.1 Procedure to Apply the Filter Step

If a multistage time discretization such as the Runge-Kutta method is desired, the spatial differencing base scheme discussed in the previous section is applied at every stage of the Runge-Kutta method. If viscous terms are present, we use the same order and type of base scheme for the viscous terms as for the inviscid terms.

There are two methods for applying the characteristic filter. Method 1 is to apply the filter at every stage of the Runge-Kutta step. Method 2 is to apply the filter at the end of the full Runge-Kutta step. For inviscid and strong shock interactions, method 1 might be more stable.

If one desires a time discretization that belongs to the class of LMMs, then the filter can be applied as a numerical dissipation vector in conjunction with the base scheme. The filter in this case is evaluated at $\hat{Q}^n$ for explicit LMMs. For implicit LMMs additional similar filters evaluated at the $n+1$ time level are involved. Alternatively, method 2 can be applied to LMMs as well. In this case, we apply the filter after the completion of the implicit time step. One can minimize flux evaluations by using the one-leg formulation of the LMMs of Dahlquist. The only non-dissipative (in time) second-order, two-time level one-leg method is the mid-point implicit method. Note that the noniterative linearized form of the midpoint implicit formula

reduces to the regular noniterative linearized trapezoidal formula.

For time marching to steady states using implicit LMMs, certain flow physics only require an explicit dissipation term. Also, the implicit operator can be different from the explicit operator. See Yee (1989) and Yee et al. (1990) for some efficient conservative linearized implicit forms.

3.2.2. Nonlinear Characteristic Filters

There are many possible candidates for the filter operator in conjunction with high order base schemes. Here, we prefer using filter operators whose grid stencils have a width similar to that of the base scheme for efficiency and ease of numerical boundary treatment. Higher than third-order filter operators are of course applicable, but they are more CPU intensive and require special treatment near boundary points for stability and accuracy. The filter operator usually consists of the product of a sensor and a nonlinear dissipation. Two possible sensors are considered, the ACM sensor (Harten 1978) and the wavelet sensor (Sjogreen & Yee 2000). See Table 1.1 for the road map. Here we briefly review the ACM sensor and interested readers are referred to Sjogreen & Yee (2000) for the wavelet sensor.

Yee et al (1999a) use nonlinear dissipation terms in conjunction with the Harten ACM sensor applied to each characteristic wave as the filter vector. In essence, the nonlinear dissipation terms act as second or third-order ACM-like operators, unlike Harten's first-order ACM (Harten 1978). The sensor is used to signal the amount of nonlinear dissipation to be added to the high order non-dissipative scheme, one wave at a time. Thus, the current approach is also different in spirit from the original Harten (1983) second-order TVD scheme which uses ACM to sharpen the contact discontinuities. Let the filter vector in the ξ -direction be of the form

$$\tilde{E}_{i+\frac{1}{2},j,k}^* = \frac{1}{2} R_{i+\frac{1}{2}} \Phi_{i+\frac{1}{2}}^* (S^\xi)_{i+\frac{1}{2},j,k}. \quad (3.2.1)$$

$\tilde{E}_{i+\frac{1}{2},j,k}^*$ is the modified form of the nonlinear dissipation portion of the standard numerical flux in curvilinear coordinates. $(S^\xi)_{i+\frac{1}{2},j,k}$ is the magnitude of the surface area vector $\mathbf{S}^\xi$ at the $(i + 1/2, j, k)$ averaged state. The quantity $R_{i+\frac{1}{2}}$ is the right eigenvector matrix given in Section 3.4.1, using, Roe's approximate solver (Roe's averaged state) with the (k, l) indices suppressed. We define $\tilde{F}_{i,j+\frac{1}{2},k}^*$ and $\tilde{G}_{i,j,k+\frac{1}{2}}^*$ in the same manner. The elements of $\Phi_{i+\frac{1}{2}}^*$, denoted by $(\phi_{i+\frac{1}{2}}^l)^*$ with the (k, l) indices suppressed, are

$$(\phi_{i+\frac{1}{2}}^l)^* = \kappa \theta_{i+\frac{1}{2}}^l \phi_{i+\frac{1}{2}}^l. \quad (3.2.2)$$

$\phi_{i+\frac{1}{2}}^l$ in (3.2.2) are the elements of the dissipative portion of the high resolution schemes resulting from using a TVD, MUSCL with slope limiters, ENO or WENO scheme. Hereafter, we refer to (3.2.2) as the ACM filter. Formulae for $\phi_{i+\frac{1}{2}}^l$ involve the eigenvalues, eigenvectors and Roe's averaged state defined in Sections 3.4.1 and 3.4.2, and can be found in the literature (Yee 1985, Yee 1989, Yee et al. 1990). See Yee et al. (1999a,b) for details and for a discussion of other possible filters.

The function $\kappa \theta_{i+\frac{1}{2}}^l$ is a mechanism to control the excess numerical dissipation inherent in the dissipative portion of high-resolution shock-capturing schemes. In other words, the

elements of $\Phi_{i+\frac{1}{2}}^*$ are the same as the nonlinear dissipation portion of the TVD, ENO or WENO scheme, multiplied by $\kappa\theta_{i+\frac{1}{2}}^l$. The parameter κ is problem dependent. For smooth flows, κ can be very small. It is used to minimize spurious high frequency oscillation producing nonlinear instability. Different physical problems may require different values of κ because of variation in flow properties. The κ value may vary from one characteristic wave to another, and from one region of the flow field to another region with a different flow structure. The range of κ for the numerical experiments in Yee et al. (1999a,b) is $0.0 \leq \kappa \leq 0.7$. The function $\theta_{i+\frac{1}{2}}^l$ is the Harten ACM sensor. For a general $2m+1$ -point base scheme, Harten recommended

$$\theta_{i+\frac{1}{2}}^l = \max(\widehat{\theta}_{i-m+1}^l, \dots, \widehat{\theta}_{i+m}^l), \quad (3.2.3)$$

$$\widehat{\theta}_i^l = \left| \frac{|\alpha_{i+\frac{1}{2}}^l| - |\alpha_{i-\frac{1}{2}}^l|}{|\alpha_{i+\frac{1}{2}}^l| + |\alpha_{i-\frac{1}{2}}^l|} \right|^p. \quad (3.2.4)$$

The $\alpha_{i+\frac{1}{2}}^l$ are elements of $R_{i+\frac{1}{2}}^{-1}(Q_{i+1,j,k} - Q_{i,j,k})$ defined in Section 3.4.1. They are used in (3.2.4), and are needed for $\phi_{i+\frac{1}{2}}^l$ in (3.2.2) as well. The parameter p in (3.2.4) is an exponent ≥ 1 and is not the “pressure p ”. The $\phi_{i+\frac{1}{2}}^l$, $\alpha_{i+\frac{1}{2}}^l$ and $R_{i+\frac{1}{2}}$ for the corresponding numerical dissipation using the MUSCL formulation are now functions of the left and right states of Q .

Instead of varying κ for the particular flow problem, one can vary p . For larger p , less numerical dissipation is added. Note that by varying $p \geq 1$ in (3.2.4), one can essentially increase the order of accuracy of the dissipation term. The order of the dissipation depends on the value of p . One can switch from $p = 1$ near shock locations to $p > 1$ at smooth regions. For all of the numerical examples in Yee et al. (1999a,b), they use $p = 1$ and

$$\theta_{i+\frac{1}{2}}^l = \max(\widehat{\theta}_i^l, \widehat{\theta}_{i+1}^l). \quad (3.2.5)$$

The shock-turbulence interaction problems appear to favor this form of $\theta_{i+\frac{1}{2}}^l$. To avoid the need to tune the arbitrary parameter κ , Sjogreen & Yee (2000) suggest replacing $\kappa\theta_{i+\frac{1}{2}}^l$ by a multi-resolution non-orthogonal wavelet sensor $\omega_{i+\frac{1}{2}}^l$. Their study shows that the wavelet sensor indeed removes the parameter dependence κ and p with comparable accuracy for the same numerical examples illustrated in Yee et al. (1999a,b). Unlike the ACM sensor, the wavelet sensor can detect most of the distinct flow features including turbulence, leading to an automatic selection of the appropriate distribution of numerical dissipation and a good grid adaptation indicator.

To avoid some conditional statements in the actual computer code and to promote vectorization, several of the functions inside the filter with the potential of dividing by zero are modified. In particular, the sensor (3.2.4), with $p = 1$, is replaced by

$$\theta_i^l = \frac{||\alpha_{i+1/2}^l| - |\alpha_{i-1/2}^l||}{|\alpha_{i+1/2}^l| + |\alpha_{i-1/2}^l| + \epsilon}. \quad (3.2.6)$$

In all of the computations, they take $\epsilon = 10^{-7}$. (Actually, ϵ should have the same dimension as $\alpha_{i+1/2}^l$.)

It is emphasized here that neither ACM nor wavelets sensors will be able to improve the accuracy at the shock and shear locations over the inherent shock-capturing capability of the nonlinear dissipation. The accuracy of the shock and shear is dictated by the chosen nonlinear dissipation. The role of the sensors is to allow the full amount of numerical dissipation in shock and shear regions, and to limit the amount of numerical dissipation in regions immediately away from shock and shear locations and the rest of the flow field. Therefore, with a suitable sensor, one does not have to use CPU-intensive high order high-resolution shock-capturing numerical dissipation, since this type of dissipation generally only gives a slightly more accurate solution away from discontinuities while exhibiting the same shock and shear resolution as the high-resolution second or third-order numerical dissipations.

Finally, we note that the ACM filter (3.2.2) or wavelet filter might not be sufficient for (a) time-marching to steady state and (b) spurious high frequency oscillations due to insufficient grid resolution and nonlinear instability away from discontinuities. See Yee et al. (1999b) for a discussion.

A Form of the Full Discretization: Here we illustrate the form of the full discretization using Runge-Kutta methods with the filters applied at the completion of a full Runge-Kutta time step. Let $\tilde{Q}^{n+1}$ be the solution after one full Runge-Kutta time step using a non-dissipative base scheme. If entropy splitting is employed, the base scheme is applied to the split form of the inviscid flux derivatives. The solution at the next time level $\hat{Q}^{n+1}$ is

$$\hat{Q}_{i,j,k}^{n+1} = \tilde{Q}_{i,j,k}^{n+1} + \frac{\Delta t}{\Delta \xi} \left[\tilde{E}_{i+\frac{1}{2},j,k}^* - \tilde{E}_{i-\frac{1}{2},j,k}^* \right] + \frac{\Delta t}{\Delta \eta} \left[\tilde{F}_{i,j+\frac{1}{2},k}^* - \tilde{F}_{i,j-\frac{1}{2},k}^* \right] + \frac{\Delta t}{\Delta \zeta} \left[\tilde{G}_{i,j,k+\frac{1}{2}}^* - \tilde{G}_{i,j,k-\frac{1}{2}}^* \right]. \quad (3.2.7)$$

Here, $\tilde{E}_{i\pm\frac{1}{2},j,k}^*$, $\tilde{F}_{i,j\pm\frac{1}{2},k}^*$ and $\tilde{G}_{i,j,k\pm\frac{1}{2}}^*$ are evaluated at $\tilde{Q}^{n+1}$.

3.3. Discretization of Geometric Terms for the Base Scheme

For the base scheme, all first partial derivatives with respect to the transformed variables, such as those in (2.3.10), are approximated by the high order compact and non-compact central difference formulas of Section 3.1. But the surface area vector components in the transformed fluxes and the cell volume in the transformed conservative *vector* are also defined in terms of partial derivatives of the Cartesian coordinates. It would be desirable for the numerical approximations to the surface area vectors to satisfy the surface area conservation law (2.3.13) exactly (within round-off errors). For time-varying grids, we would also like the volume conservation law (2.3.14) to be satisfied numerically. This would eliminate numerical errors in regions of uniform flow, and reduce grid induced errors even for nonuniform flows. Furthermore, in deriving the split formula (2.3.10), (2.3.13) was used in obtaining the expression for the second term.

It was shown by Vinokur (1989) that the two geometric conservation laws can be satisfied numerically to second-order accuracy by appealing to finite-volume concepts. Unfortunately,

this approach cannot be extended to higher orders of accuracy. Calculating the surface area vectors by applying the high order difference formulas of Section 3.1 to (2.3.2) will result in truncation errors when evaluating (2.3.13). Gaitonde & Visbal (1999) showed, by performing some numerical experiments, that (2.3.13) can be satisfied numerically by rewriting the component version of (2.3.2) in an equivalent “conservative” form first proposed by Thomas & Lombard (1979). That form is not coordinate invariant. In Section 3.3.1 we present a coordinate invariant form of the Thomas and Lombard relations for three-dimensional flow. We then derive the corresponding correct relations for axis-symmetric flow in Section 3.3.2. This cannot be accomplished using the original Thomas and Lombard formulas. We will also demonstrate theoretically in Appendix A that applying any arbitrary compact or non-compact difference formulas to the new form of the surface area vector definition (2.3.2) will satisfy the surface area conservation law (2.3.13) exactly.

3.3.1. Three-Dimensional Flow

By examining the Thomas and Lombard formulas, one can readily obtain the coordinate invariant form for $\mathbf{S}^\xi$ as

$$\mathbf{S}^\xi = \frac{1}{2}[(\mathbf{r}_\eta \times \mathbf{r})_\zeta - (\mathbf{r}_\zeta \times \mathbf{r})_\eta], \quad (3.3.1)$$

with the corresponding expressions for $\mathbf{S}^\eta$ obtained by replacing η and ζ by ζ and ξ , respectively, and for $\mathbf{S}^\zeta$ obtained by replacing η and ζ by ξ and η , respectively. Analytically, (3.3.1) gives the same answer as provided by (2.3.2), noting that since the mixed partial derivatives commute, $\mathbf{r}_{\eta\zeta} \times \mathbf{r} - \mathbf{r}_{\zeta\eta} \times \mathbf{r} = 0$. The commuting property also assures that the answer for $\mathbf{S}^\xi$ is independent of the choice of origin for $\mathbf{r}$, and that (2.3.13) is identically satisfied. In order to show that (2.3.13) is satisfied exactly numerically, we must prove that the mixed partial derivatives commute **numerically**. This we do in Appendix A.

In component form, (3.3.1) is written as

$$S_1^\xi = \frac{1}{2}[(y_\eta z - z_\eta y)_\zeta - (y_\zeta z - z_\zeta y)_\eta], \quad (3.3.2a)$$

$$S_2^\xi = \frac{1}{2}[(z_\eta x - x_\eta z)_\zeta - (z_\zeta x - x_\zeta z)_\eta], \quad (3.3.2b)$$

$$S_3^\xi = \frac{1}{2}[(x_\eta y - y_\eta x)_\zeta - (x_\zeta y - y_\zeta x)_\eta]. \quad (3.3.2c)$$

Expressions for components of $\mathbf{S}^\eta$ and $\mathbf{S}^\zeta$ are obtained in an analogous manner. The Thomas and Lombard forms used twice the first term in each parenthesis. While this satisfies (2.3.13), it will not give the correct formulas for axis-symmetric flow, as will be discussed in Section 3.3.2.

From (2.3.7) we see that the transformed grid velocities involve products of surface area components and grid velocity components, which cannot be expressed in “conservative” form. Consequently, we cannot express $\mathcal{V}$ numerically so as to satisfy (2.3.14) exactly, when the surface area components are given by (3.3.2). We therefore calculate $\mathcal{V}$ from the invariant expression (2.3.3) as

$$\mathcal{V} = \begin{vmatrix} x_\xi & y_\xi & z_\xi \\ x_\eta & y_\eta & z_\eta \\ x_\zeta & y_\zeta & z_\zeta \end{vmatrix}. \quad (3.3.3)$$

3.3.2 Axi-Symmetric Flow

The axi-symmetric surface area components are obtained by substituting (2.3.16) and (2.3.17) into (3.3.2), and then setting $\zeta = 0$. From (2.3.17) it follows that $z = 0$ and r can be replaced by y . The resulting expressions are

$$S_1^\xi = \frac{1}{2}(y^2)_\eta \quad S_2^\xi = \frac{1}{2}[y_\eta x - x_\eta y - (xy)_\eta] \quad S_3^\xi = 0, \quad (3.3.4a)$$

$$S_1^\eta = -\frac{1}{2}(y^2)_\xi \quad S_2^\eta = -\frac{1}{2}[y_\xi x - x_\xi y - (xy)_\xi] \quad S_3^\eta = 0, \quad (3.3.4b)$$

$$S_1^\zeta = 0 \quad S_2^\zeta = 0 \quad S_3^\zeta = S^\zeta, \quad (3.3.4c)$$

where

$$S^\zeta = \frac{1}{2}[(x_\xi y - y_\xi x)_\eta - (x_\eta y - y_\eta x)_\xi]. \quad (3.3.5)$$

We also have the relations

$$(S_1^\zeta)_\zeta = 0 \quad (S_2^\zeta)_\zeta = -S^\zeta \quad (S_3^\zeta)_\zeta = 0. \quad (3.3.6)$$

Substituting (3.3.4), (3.3.5), and (3.3.6) into (2.3.13), we see that the surface area conservation law is satisfied. The cell volume is given by

$$\mathcal{V} = yS^\zeta. \quad (3.3.7)$$

If we used the Thomas and Lombard form of (3.3.2), the resulting equations (3.3.4) to (3.3.6) would have different expressions. While they would still satisfy (2.3.13), they would lead to numerical errors which violate the requirements of axial symmetry. This is a direct result of the lack of coordinate invariance in their form. Equations (3.3.4c) would still hold, with (3.3.5) replaced by twice the first term in each parenthesis; that is

$$S^\zeta = (x_\xi y)_\eta - (x_\eta y)_\xi. \quad (3.3.8)$$

The expressions for their ζ derivatives would not be physically correct. The first two would now be

$$(S_1^\zeta)_\zeta = (yy_\xi)_\eta - (yy_\eta)_\xi \quad (S_2^\zeta)_\zeta = (xy_\xi)_\eta - (xy_\eta)_\xi. \quad (3.3.9)$$

Analytically, $(S_1^\zeta)_\zeta = 0$ and $(S_2^\zeta)_\zeta = -S^\zeta$, corresponding to the fact that the constant ζ surfaces are planes parallel to the x -axis. But numerically, these two conditions are only satisfied to the order of the truncation error of the method. For both these reasons, this would result in small contributions to the source term $\hat{G}_\zeta$ in both the continuity, x -momentum and energy equations. By using the coordinate invariant expressions (3.3.2), we satisfy (3.3.6) exactly, and avoid these spurious numerical results.

3.4. Riemann Solver

3.4.1. Flux Jacobians and Eigenvectors

Characteristic based filters require the use of eigenvalues and eigenvectors of flux Jacobian matrices at averaged states. They are best treated using the original conservative *vector* Q rather than $\hat{Q}$. Let $\mathbf{S}^\xi = S^\xi \mathbf{n}^\xi$, where $\mathbf{n}^\xi$ is the unit normal and S^ξ is the magnitude of the surface area vector $\mathbf{S}^\xi$. Similar definitions apply to $\mathbf{S}^\eta$ and $\mathbf{S}^\zeta$. We omit the curvilinear coordinate superscript from now on. At any averaged state, only the direction of the appropriate normal $\mathbf{n} = n_1 \mathbf{i} + n_2 \mathbf{j} + n_3 \mathbf{k}$ plays a role. Let $u^n = \mathbf{u} \cdot \mathbf{n}$, $v^n = \mathbf{v} \cdot \mathbf{n}$, and $u' = u^n - v^n$. The transformed flux *vector* can be written as $\mathbf{S} \cdot \mathbf{F} = SF^n$, where $F^n = \mathbf{n} \cdot \mathbf{F}$. The flux Jacobian matrix is then defined as $A = (F^n)_Q = \partial F^n / \partial Q$.

We will first develop relations for a general equilibrium gas. The pressure p is given by a general equation of state of the form

$$p = p(\rho, \tilde{\epsilon}). \quad (3.4.1)$$

The derivatives will be denoted by

$$\chi = \left(\frac{\partial p}{\partial \rho} \right)_{\tilde{\epsilon}} \quad \text{and} \quad \kappa = \left(\frac{\partial p}{\partial \tilde{\epsilon}} \right)_{\rho}. \quad (3.4.2)$$

If $h = (\tilde{\epsilon} + p)/\rho$ is the specific enthalpy, the speed of sound c can then be expressed as

$$c^2 = \chi + \kappa h. \quad (3.4.3)$$

The matrix A can then be written as

$$A = \begin{bmatrix} -v^n & n_1 & n_2 & n_3 & 0 \\ Kn_1 - u^n u & (1 - \kappa)un_1 + u' & un_2 - \kappa n_1 v & un_3 - \kappa n_1 w & \kappa n_1 \\ Kn_2 - u^n v & vn_1 - \kappa n_2 u & (1 - \kappa)vn_2 + u' & vn_3 - \kappa n_2 w & \kappa n_2 \\ Kn_3 - u^n w & wn_1 - \kappa n_3 u & wn_2 - \kappa n_3 v & (1 - \kappa)wn_3 + u' & \kappa n_3 \\ (K - H)u^n & Hn_1 - \kappa u^n u & Hn_2 - \kappa u^n v & Hn_3 - \kappa u^n w & u' + \kappa u^n \end{bmatrix}, \quad (3.4.4)$$

where $K = \frac{1}{2}\kappa q^2 + \chi$, and $H = h + \frac{1}{2}q^2$ is the total enthalpy per unit mass. The three distinct eigenvalues of A are

$$\lambda^1 = u', \quad \lambda^2 = u' + c, \quad \text{and} \quad \lambda^3 = u' - c, \quad (3.4.5)$$

where λ^1 is a repeated eigenvalue. In order to obtain a corresponding set of linearly independent eigenvectors, we introduce an arbitrary orthonormal basis $\mathbf{a}^i$ satisfying $\mathbf{a}^i \cdot \mathbf{a}^j = \delta^{ij}$, where δ^{ij} is the Kronecker delta. One can then define $a^{ni} = \mathbf{n} \cdot \mathbf{a}^i$ and $\mathbf{b}^i = \mathbf{n} \times \mathbf{a}^i$. The right eigenvector matrix R can then be written as

$$R = \begin{bmatrix} a^{n1} & a^{n2} & a^{n3} & 1 & 1 \\ a^{n1}u + cb_1^1 & a^{n2}u + cb_1^2 & a^{n3}u + cb_1^3 & u + cn_1 & u - cn_1 \\ a^{n1}v + cb_2^1 & a^{n2}v + cb_2^2 & a^{n3}v + cb_2^3 & v + cn_2 & v - cn_2 \\ a^{n1}w + cb_3^1 & a^{n2}w + cb_3^2 & a^{n3}w + cb_3^3 & w + cn_3 & w - cn_3 \\ a^{n1}\tilde{K} + c(\mathbf{b}^1 \cdot \mathbf{u}) & a^{n2}\tilde{K} + c(\mathbf{b}^2 \cdot \mathbf{u}) & a^{n3}\tilde{K} + c(\mathbf{b}^3 \cdot \mathbf{u}) & H + cu^n & H - cu^n \end{bmatrix}, \quad (3.4.6)$$

where $\tilde{K} = \frac{1}{2}q^2 - \chi/\kappa$, $\mathbf{b}^i = b_1^i \mathbf{i} + b_2^i \mathbf{j} + b_3^i \mathbf{k}$, and $\mathbf{b}^i \cdot \mathbf{u} = b_1^i u + b_2^i v + b_3^i w$. The quantity $\alpha = R^{-1}\Delta Q$ is expressed most simply as

$$\alpha = \begin{bmatrix} a^{n1}(\Delta\rho - \Delta p/c^2) + \rho(b_1^1\Delta u + b_2^1\Delta v + b_3^1\Delta w)/c \\ a^{n2}(\Delta\rho - \Delta p/c^2) + \rho(b_1^2\Delta u + b_2^2\Delta v + b_3^2\Delta w)/c \\ a^{n3}(\Delta\rho - \Delta p/c^2) + \rho(b_1^3\Delta u + b_2^3\Delta v + b_3^3\Delta w)/c \\ \frac{1}{2}(\Delta p/c^2 + \rho[n_1\Delta u + n_2\Delta v + n_3\Delta w]/c) \\ \frac{1}{2}(\Delta p/c^2 - \rho[n_1\Delta u + n_2\Delta v + n_3\Delta w]/c) \end{bmatrix}. \quad (3.4.7)$$

Note that in computing the filters, we need α but not R^{-1} explicitly. For efficiency, one should use 3.4.7 instead of doing the actual product of the matrix R^{-1} with the *vector* ΔQ , especially for nonequilibrium flows (Yee & Shinn 1989, Yee 1989).

There are three methods of calculating the a^{ni} and $\mathbf{b}^i$. They all require the knowledge of $\mathbf{n}$ at each averaged state, obtained by calculating $\mathbf{S} = S\mathbf{n}$ as the vectorial average of the $\mathbf{S}$ at the two neighboring grid points. This is consistent with the accuracy of the filter procedure. Each method has certain disadvantages. If storage is not an issue, then the third method is preferred. In the first method, the Cartesian unit vectors are used as the basis $\mathbf{a}^i$. We thus define

$$\mathbf{a}^1 = \mathbf{i}, \quad \mathbf{a}^2 = \mathbf{j}, \quad \mathbf{a}^3 = \mathbf{k}. \quad (3.4.8)$$

It then follows that

$$a^{n1} = n_1, \quad a^{n2} = n_2, \quad a^{n3} = n_3, \quad (3.4.9)$$

and

$$\begin{aligned} b_1^1 &= 0 & b_2^1 &= n_3 & b_3^1 &= -n_2 \\ b_1^2 &= -n_3 & b_2^2 &= 0 & b_3^2 &= n_1 \\ b_1^3 &= n_2 & b_2^3 &= -n_1 & b_3^3 &= 0. \end{aligned} \quad (3.4.10)$$

The disadvantage of this method is that both R and $R^{-1}\Delta Q$ involve many terms to be evaluated. A number of these terms would be absent if the normal $\mathbf{n}$ was one member of the basis $\mathbf{a}^i$. This choice underlies the other two methods. We could then define

$$\mathbf{a}^1 = \mathbf{n}, \quad \mathbf{a}^2 = \mathbf{t}, \quad \mathbf{a}^3 = \mathbf{s}, \quad (3.4.11)$$

where $\mathbf{n}$, $\mathbf{t}$, $\mathbf{s}$ form an orthonormal triad, with $\mathbf{t}$ as yet undefined, but satisfying $\mathbf{n} \cdot \mathbf{t} = 0$, and $\mathbf{s} = \mathbf{n} \times \mathbf{t}$. With this choice we then have

$$a^{n1} = 1, \quad a^{n2} = 0, \quad a^{n3} = 0, \quad (3.4.12)$$

and

$$\begin{aligned} b_1^1 &= 0 & b_2^1 &= 0 & b_3^1 &= 0 \\ b_1^2 &= s_1 & b_2^2 &= s_2 & b_3^2 &= s_3 \\ b_1^3 &= -t_1 & b_2^3 &= -t_2 & b_3^3 &= -t_3. \end{aligned} \quad (3.4.13)$$

The best way to define $\mathbf{t}$ is as

$$\mathbf{t} = \frac{\mathbf{c} \times \mathbf{n}}{|\mathbf{c} \times \mathbf{n}|}, \quad (3.4.14)$$

where $\mathbf{c}$ is some arbitrary vector. The choice of $\mathbf{c}$ differentiates the other two methods. In the second method, $\mathbf{c}$ is taken as a constant. A common choice is to let

$$\mathbf{c} = \mathbf{i} + \mathbf{j} + \mathbf{k}. \quad (3.4.15)$$

This involves a small number of additional operations, if storage is a problem, and $\mathbf{t}$ and $\mathbf{s}$ must be recalculated at each time step. The disadvantage of this method is that it breaks down when $\mathbf{c}$ and $\mathbf{n}$ are parallel. To avoid this eventuality, in the third method, $\mathbf{c}$ is chosen to be one of the vectors $\mathbf{S}$ at a neighboring grid point. This method involves a few more operations than the second method, but is more robust. As an example, for a ξ surface at $(i + 1/2, j, k)$, we can let $\mathbf{c}$ be $\mathbf{S}^\zeta$ at (i, j, k) .

3.4.2. Roe Average

Among the various approximate Riemann solvers, the most common one uses the Roe average because of its simplicity and its ability to satisfy the jump conditions across discontinuities exactly. In those solvers based on local linearization, the flux at a point separating two states Q^L and Q^R is based on the eigenvalues and eigenvectors of some average $\bar{A}$. The optimum choice for $\bar{A}$ is one satisfying

$$\Delta F^n = \bar{A} \Delta Q, \quad (3.4.16)$$

where $\Delta(\cdot) = (\cdot)^R - (\cdot)^L$. This choice of $\bar{A}$ captures discontinuities exactly. One way of obtaining $\bar{A}$ is to seek an average state $\bar{Q}$, which is a function of Q^L and Q^R , such that

$$\bar{A} = A(\bar{Q}). \quad (3.4.17)$$

Such a state is known as a Roe-averaged state. One can easily show that

$$\bar{\mathbf{u}} = \nu \mathbf{u}^L + (1 - \nu) \mathbf{u}^R \quad (3.4.18)$$

and

$$\bar{H} = \nu H^L + (1 - \nu) H^R, \quad (3.4.19)$$

where

$$\nu = \frac{1}{1 + \sqrt{\rho^R/\rho^L}}. \quad (3.4.20)$$

From the definition of H one then obtains

$$\bar{h} = \bar{H} - \frac{1}{2} \bar{\mathbf{u}} \cdot \bar{\mathbf{u}}. \quad (3.4.21)$$

In (3.4.7) we also need $\bar{\rho}$, which is given by

$$\bar{\rho} = \sqrt{\rho^R \rho^L}. \quad (3.4.22)$$

We still need to evaluate $\bar{\chi}$ and $\bar{\kappa}$, which are used to determine $\bar{c}$ from (3.4.3) and $\bar{K}$ from its definition. For a general equilibrium gas, various approximate methods of calculating $\bar{\chi}$ and $\bar{\kappa}$ have been proposed. Since the entropy splitting is only valid for a thermally perfect gas, we consider only this case, for which there is an exact solution. The details are found in Yee et al. (1999b). The exact formulas are

$$\bar{\kappa} = \frac{\Delta \hat{T}}{\Delta \epsilon} \quad (3.4.23a)$$

and

$$\bar{\chi} = \frac{\epsilon^R \hat{T}^L - \epsilon^L \hat{T}^R}{\Delta \epsilon}. \quad (3.4.23b)$$

Equations (3.4.23a,b) are replaced by

$$\bar{\kappa} = 1/\dot{\epsilon} \quad (3.4.24a)$$

and

$$\bar{\chi} = \hat{T} - \bar{\kappa} \epsilon \quad (3.4.24b)$$

when $\Delta\epsilon \rightarrow 0$.

For a perfect gas (3.4.23a,b) reduce to

$$\bar{\chi} = 0 \quad \text{and} \quad \bar{\kappa} = \gamma - 1. \quad (3.4.25)$$

In order to obtain $R_{i+\frac{1}{2}}$ and $\Phi_{i+\frac{1}{2}}$ for the filter (3.2.1), the right state and left state (super-scripts R and L in (3.4.16) - (3.4.23)) are the grid indices $(i+1, j, k)$ and (i, j, k) .

3.4.3. Non-equilibrium Flow

In Yee et al. (1999b), we showed that the extension of entropy splitting to fully-coupled non-equilibrium flow is not practically feasible. But the schemes in question are usable, and in addition, one can obtain an exact extension of the Roe's Riemann solver for non-equilibrium flow. This is presented in Appendix B.

Concluding Remarks

In this paper we formulate the Yee et al. (1999a) scheme with entropy splitting (Yee et al. 1999b) in 3-D curvilinear moving grids for a thermally perfect gas. For efficiency, we choose the finite difference formulation. The surface area conservation law is satisfied numerically by discretizing a coordinate invariant version of the Thomas and Lombard formulas. This form was independently proposed by the authors in June 2000, and by Thomas & Neier (1990) (P.D. Thomas, private communication, November 2000). Although the formal extension of entropy splitting is limited to a thermally perfect gas, the numerical schemes themselves do not have this restriction. Consequently, the schemes discussed here are applicable to equilibrium real gas, non-equilibrium and artificial compressibility method of treating incompressible flows, MHD and the Maxwell equations. In addition, the dual purpose wavelet sensors (dynamic numerical dissipation controls and grid adaptation indicators) proposed by Sjogreen and Yee can be a stand alone option for a variety of schemes other than what is discussed here. Numerical experiments with the metric terms in general coordinate transformation that are discretized by the same high order difference operator as the flow variables can be found in Müller & Yee (2001) and Polifke et al. (2001). Numerical examples illustrating the performance of the new 3-D metric formulation will be reported in a future paper.

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APPENDIX A

The Commutativity of a Class of Numerical Mixed Partial Derivatives

In this appendix we prove that the numerical mixed partial derivatives commute, so that the surface area conservation law is satisfied exactly. We would like to thank Dennis Jespersen of NASA Ames Research Center for providing the essential elements of the proof. We find it convenient to employ a notation which differs from that in the body of the paper. Upper

case letters denote a matrix, lower case letters denote an algebraic vector, and Latin subscripts denote their components.

We first introduce the notion of a tensor product (or Kronecker product). Given two arbitrary matrices A and B , the tensor product $A \otimes B$ is defined as a block matrix, each of whose components is the matrix B multiplied by the corresponding component of the matrix A . In component form this is written as

$$(A \otimes B)_{ij} = A_{ij}B. \quad (\text{A.1})$$

From the definition (A.1), we can derive an important mixed product rule. If A and C are conformable matrices, and B and D are also conformable, then

$$(A \otimes B)(C \otimes D) = AC \otimes BD. \quad (\text{A.2})$$

The proof follows immediately by writing (A.2) in component form. Thus

$$\begin{aligned} [(A \otimes B)(C \otimes D)]_{ik} &= \sum_j (A \otimes B)_{ij} (C \otimes D)_{jk} \\ &= \sum_j A_{ij} B C_{jk} D \\ &= \left(\sum_j A_{ij} C_{jk} \right) B D \\ &= (AC)_{ik} B D \\ &= (AC \otimes B D)_{ik}. \end{aligned} \quad (\text{A.3})$$

Let the ξ, η, ζ computational space be discretized with l, m, n points in the ξ, η, ζ directions, respectively. For a fixed η and ζ , the most general finite-difference approximation of the ξ derivative is

$$A^\xi u_\xi = B^\xi u, \quad (\text{A.4})$$

where u and u_ξ are l -dimensional vectors, and A^ξ and B^ξ are l by l matrices. Some examples are the central non-compact and compact spatial schemes (3.1.1) - (3.1.3). No restrictions are placed on the nature of A^ξ and B^ξ , which incorporate arbitrary boundary conditions on the ξ boundaries of the computational region. Assume that the discrete unknowns for the whole region are ordered with ξ values varying first, η values varying next, and ζ values varying last. If the **same** finite-difference approximation (A.4) is applied for each η and ζ (which implies the same boundary condition along each of the ξ boundaries), then the approximation to the ξ derivatives of all the unknowns is

$$\overline{A}^\xi \overline{u}_\xi = \overline{B}^\xi \overline{u}, \quad (\text{A.5})$$

where $\overline{u}$ and $\overline{u}_\xi$ are $l \times m \times n$ -dimensional vectors, and $\overline{A}^\xi$ and $\overline{B}^\xi$ are $l \times m \times n$ by $l \times m \times n$ matrices given by

$$\overline{A}^\xi = I^n \otimes (I^m \otimes A^\xi), \quad \overline{B}^\xi = I^n \otimes (I^m \otimes B^\xi). \quad (\text{A.6})$$

I^n and I^m are the n by n and m by m identity matrices, respectively. Note that the parentheses in (A.6) can be eliminated, since from its definition, tensor multiplication can be shown to be associative.

For a fixed ξ and ζ , the finite-difference approximation of the η derivative can be written as

$$A^\eta v_\eta = B^\eta v, \quad (\text{A.7})$$

where v and v_η are m -dimensional vectors, and A^η and B^η are m by m matrices. Note that A^η and B^η are again arbitrary, with different boundary conditions on the η boundaries than on the ξ boundaries being permitted. If the same finite-difference approximation (A.7) is applied for each ξ and ζ , then the $l \times m \times n$ -dimensional vectors $\bar{v}$ and $\bar{v}_\eta$ of all the unknowns are related by

$$\bar{A}^\eta \bar{v}_\eta = \bar{B}^\eta \bar{v}, \quad (\text{A.8})$$

where the $l \times m \times n$ by $l \times m \times n$ matrices $\bar{A}^\eta$ and $\bar{B}^\eta$ are given by

$$\bar{A}^\eta = I^n \otimes (A^\eta \otimes I^l), \quad \bar{B}^\eta = I^n \otimes (B^\eta \otimes I^l). \quad (\text{A.9})$$

I^l is the l by l identity matrix. Similarly, for a fixed ξ and η , the finite-difference approximation of the ζ derivative can be written as

$$A^\zeta w_\zeta = B^\zeta w, \quad (\text{A.10})$$

where w and w_ζ are n -dimensional vectors, and A^ζ and B^ζ are n by n matrices. A^ζ and B^ζ are again arbitrary, with different boundary conditions on the ζ boundaries than on the other boundaries being permitted. If the same finite-difference approximation (A.10) is applied for each ξ and η , then the $l \times m \times n$ -dimensional vectors $\bar{w}$ and $\bar{w}_\zeta$ of all the unknowns are related by

$$\bar{A}^\zeta \bar{w}_\zeta = \bar{B}^\zeta \bar{w}, \quad (\text{A.11})$$

where the $l \times m \times n$ by $l \times m \times n$ matrices $\bar{A}^\zeta$ and $\bar{B}^\zeta$ are given by

$$\bar{A}^\zeta = A^\zeta \otimes (I^m \otimes I^l), \quad \bar{B}^\zeta = B^\zeta \otimes (I^m \otimes I^l). \quad (\text{A.12})$$

Let the vector $\bar{u}_{\xi\eta}$ be the finite-difference approximation of all the mixed $\xi\eta$ -partial derivatives, calculated by first obtaining $\bar{u}_\xi$ using (A.5), and then using (A.8) to obtain $\bar{u}_{\xi\eta}$. If the grid points are equally spaced in the computational space, then the matrices $A^\xi, B^\xi, A^\eta, B^\eta, A^\zeta$ and B^ζ are all **constant**. The final result then becomes

$$\bar{A}^\eta (\bar{A}^\xi \bar{u}_\xi)_\eta = \bar{A}^\eta \bar{A}^\xi \bar{u}_{\xi\eta} = \bar{B}^\eta \bar{B}^\xi \bar{u}. \quad (\text{A.13})$$

If we carry out the operations in the reverse order, we obtain

$$\bar{A}^\xi \bar{A}^\eta \bar{u}_{\eta\xi} = \bar{B}^\xi \bar{B}^\eta \bar{u}. \quad (\text{A.14})$$

Using (A.2), (A.6) and (A.9), it follows that

$$\begin{aligned} \bar{A}^\eta \bar{A}^\xi &= [I^n \otimes (A^\eta \otimes I^l)] [I^n \otimes (I^m \otimes A^\xi)] \\ &= I^n \otimes [(A^\eta \otimes I^l)(I^m \otimes A^\xi)] \\ &= I^n \otimes (A^\eta \otimes A^\xi). \end{aligned} \quad (\text{A.15})$$

In the same manner we show that

$$\begin{aligned}\overline{A}^\xi \overline{A}^\eta &= [I^n \otimes (I^m \otimes A^\xi)][I^n \otimes (A^\eta \otimes I^l)] \\ &= I^n \otimes [(I^m \otimes A^\xi)(A^\eta \otimes I^l)] \\ &= I^n \otimes (A^\eta \otimes A^\xi).\end{aligned}\tag{A.16}$$

Therefore from (A.15) and (A.16) we have established that

$$\overline{A}^\eta \overline{A}^\xi = \overline{A}^\xi \overline{A}^\eta.\tag{A.17}$$

In a similar manner we show that

$$\overline{B}^\eta \overline{B}^\xi = \overline{B}^\xi \overline{B}^\eta.\tag{A.18}$$

From (A.13) and (A.14) it follows that

$$\overline{u}_{\xi\eta} = \overline{u}_{\eta\xi}.\tag{A.19}$$

In a similar manner, using (A.5) and (A.11), the finite-difference approximation of all the mixed $\xi\zeta$ -partial derivatives can be shown to satisfy

$$\overline{A}^\zeta (\overline{A}^\xi \overline{u}_\xi)_\zeta = \overline{A}^\zeta \overline{A}^\xi \overline{u}_{\xi\zeta} = \overline{B}^\zeta \overline{B}^\xi \overline{u}.\tag{A.20}$$

If we carry out the operations in the reverse order, we obtain

$$\overline{A}^\xi \overline{A}^\zeta \overline{u}_{\zeta\xi} = \overline{B}^\xi \overline{B}^\zeta \overline{u}.\tag{A.21}$$

Using (A.2), (A.6) and (A.12), it follows that

$$\overline{A}^\xi \overline{A}^\zeta = A^\zeta \otimes (I^m \otimes A^\xi) = \overline{A}^\zeta \overline{A}^\xi,\tag{A.22}$$

with an analogous relation involving $\overline{B}^\xi$ and $\overline{B}^\zeta$. From (A.20) and (A.21) it then follows that

$$\overline{u}_{\xi\zeta} = \overline{u}_{\zeta\xi}.\tag{A.23}$$

The same procedure, using (A.8) and (A.11), can be used to show that

$$\overline{u}_{\eta\zeta} = \overline{u}_{\zeta\eta}.\tag{A.24}$$

This completes the proof that all numerical mixed partial derivatives commute.

APPENDIX B

Riemann Solver for Non-equilibrium Flow

In this appendix we present the Riemann solver for chemical non-equilibrium flow. The analysis can be readily extended to thermo-chemical nonequilibrium. We consider a mixture of ns chemical species, each obeying a thermally perfect gas law. Let ρ^s , R^s , and $\epsilon^s(T)$ be the density, gas constant, and internal energy for species s , respectively. T is the temperature

of the mixture, assumed to be in thermal equilibrium. Note that we have not introduced a normalized $\hat{T}$, since the R^s are different for each species. All species satisfy the condition $\dot{\epsilon}^s > 0$, where $\dot{\epsilon}^s = d\epsilon^s/dT$. We can now define the density of the mixture, ρ , as

$$\rho = \sum_s \rho^s, \quad (\text{B.1})$$

the species mass fraction α^s as

$$\alpha^s \equiv \frac{\rho^s}{\rho}, \quad (\text{B.2})$$

the pressure p as

$$p = \rho RT, \quad (\text{B.3})$$

where

$$\rho R = \sum_s \rho^s R^s, \quad (\text{B.4})$$

and the internal energy of the mixture per unit volume as

$$\tilde{\epsilon} = \rho \epsilon = \sum_s \rho^s \epsilon^s. \quad (\text{B.5})$$

The conservation law takes the form

$$Q_t + \nabla \cdot \mathbf{F} = \mathcal{S}^c, \quad (\text{B.6})$$

where $\mathcal{S}^c$ is a *vector* consisting of source terms for each species. The *vectors* Q and $\mathbf{F}$ take the form

$$Q = \begin{bmatrix} (\rho^s) \\ \rho u \\ \rho v \\ \rho w \\ e \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} (\rho^s \mathbf{u}') \\ \rho u \mathbf{u}' + p \mathbf{i} \\ \rho v \mathbf{u}' + p \mathbf{j} \\ \rho w \mathbf{u}' + p \mathbf{k} \\ (e + p) \mathbf{u}' + p \mathbf{v} \end{bmatrix}. \quad (\text{B.7})$$

Note that $\mathbf{F}$ is again defined relative to a moving grid. Q and $\mathbf{F}$ are $(ns + 4)$ -dimensional *vectors*. We employ compact notations “ (ρ^s) ” and “ $(\rho^s \mathbf{u}')$ ” in which the first term written for each *vector* in (B.7) is a general component of a ns -dimensional *subvector* corresponding to the ns species continuity equations. The temperature $T(Q)$ is obtained by solving implicitly the equation

$$\sum_s \rho^s \epsilon^s(T) = e - \frac{1}{2\rho}[(\rho u)^2 + (\rho v)^2 + (\rho w)^2]. \quad (\text{B.8})$$

Equation (B.8) has a unique solution since $\rho^s > 0$ and $\dot{\epsilon}^s > 0$.

In order to derive the flux Jacobian matrix, the pressure p must be expressed in the form

$$p = p(\rho^s, \tilde{\epsilon}). \quad (\text{B.9})$$

The derivatives will be denoted by

$$\chi^s = \left(\frac{\partial p}{\partial \rho^s} \right)_{\tilde{\epsilon}, \rho^r} \quad \text{and} \quad \kappa = \left(\frac{\partial p}{\partial \tilde{\epsilon}} \right)_{\rho^s}. \quad (\text{B.10})$$

With the aid of (B.1) to (B.5) we can show that

$$\kappa = \frac{\sum_s \alpha^s R^s}{\sum_s \alpha^s \dot{\epsilon}^s} \quad (\text{B.11})$$

and

$$\chi^s = R^s T - \kappa \epsilon^s. \quad (\text{B.12})$$

The matrix A can then be written as

$$A = \begin{bmatrix} (\delta^{sr} u' - \alpha^s u^n) & (\alpha^s n_1) & (\alpha^s n_2) & (\alpha^s n_3) & 0 \\ (K^r n_1 - u^n u) & (1 - \kappa) u n_1 + u' & u n_2 - \kappa n_1 v & u n_3 - \kappa n_1 w & \kappa n_1 \\ (K^r n_2 - u^n v) & v n_1 - \kappa n_2 u & (1 - \kappa) v n_2 + u' & v n_3 - \kappa n_2 w & \kappa n_2 \\ (K^r n_3 - u^n w) & w n_1 - \kappa n_3 u & w n_2 - \kappa n_3 v & (1 - \kappa) w n_3 + u' & \kappa n_3 \\ ((K^r - H) u^n) & H n_1 - \kappa u^n u & H n_2 - \kappa u^n v & H n_3 - \kappa u^n w & u' + \kappa u^n \end{bmatrix}, \quad (\text{B.13})$$

where $K^r = \frac{1}{2} \kappa q^2 + \chi^r$ and δ^{sr} is the Kronecker delta. All other quantities are defined in Section 3.4. Once again a compact notation is used for the terms written in the first row of A . The first term written in A is the sr component of a ns by ns submatrix, and the second to fourth terms are column subvectors of dimension ns . In addition, the last four terms written in the first column of A are row subvectors of dimension ns .

The three distinct eigenvalues of A are again

$$\lambda^1 = u', \quad \lambda^2 = u' + c, \quad \text{and} \quad \lambda^3 = u' - c, \quad (\text{B.14})$$

where c is the frozen speed of sound given by

$$c^2 = \sum_s \alpha^s \chi^s + \kappa h. \quad (\text{B.15})$$

λ^1 is the repeated eigenvalue of multiplicity $ns + 2$. The choice of Cartesian unit vectors (3.4.8) as the basis $\mathbf{a}^i$ in order to obtain the set of linearly independent eigenvectors would result in far too many additional terms in R and $R^{-1} \Delta Q$ for non-equilibrium flow. We therefore restrict ourselves to the triad $\mathbf{n}, \mathbf{t}, \mathbf{s}$ as the basis (3.4.11). The right eigenvector matrix R can then be written as

$$R = \begin{bmatrix} (\delta^{sr}) & 0 & 0 & (\alpha^s) & (\alpha^s) \\ u & cs_1 & -ct_1 & u + cn_1 & u - cn_1 \\ v & cs_2 & -ct_2 & v + cn_2 & v - cn_2 \\ w & cs_3 & -ct_3 & w + cn_3 & w - cn_3 \\ \tilde{K}^r & c(\mathbf{s} \cdot \mathbf{u}) & -c(\mathbf{t} \cdot \mathbf{u}) & H + cu^n & H - cu^n \end{bmatrix}, \quad (\text{B.16})$$

where $\tilde{K}^r = \frac{1}{2} q^2 - \chi^r / \kappa$. The quantity $R^{-1} \Delta Q$ is expressed most simply as

$$R^{-1} \Delta Q = \begin{bmatrix} (\Delta \rho^s - \alpha^s \Delta p / c^2) \\ \rho(s_1 \Delta u + s_2 \Delta v + s_3 \Delta w) / c \\ -\rho(t_1 \Delta u + t_2 \Delta v + t_3 \Delta w) / c \\ \frac{1}{2} (\Delta p / c^2 + \rho[n_1 \Delta u + n_2 \Delta v + n_3 \Delta w] / c) \\ \frac{1}{2} (\Delta p / c^2 - \rho[n_1 \Delta u + n_2 \Delta v + n_3 \Delta w] / c) \end{bmatrix}. \quad (\text{B.17})$$

Equations (3.4.16) to (3.4.22) defining Roe-averaged variables are still valid. In addition, we define

$$\overline{T} = \nu T^L + (1 - \nu) T^R, \quad (\text{B.18})$$

$$\overline{\alpha^s} = \nu \alpha^{sL} + (1 - \nu) \alpha^{sR}, \quad (\text{B.19})$$

and

$$\overline{\epsilon^s} = \nu \epsilon^{sL} + (1 - \nu) \epsilon^{sR}. \quad (\text{B.20})$$

We can then show that (3.4.16) is satisfied if we define

$$\overline{\kappa} = \left(\frac{\sum_s \overline{\alpha^s} R^s}{\sum_s \overline{\alpha^s} \Delta \epsilon^s} \right) \Delta T, \quad (\text{B.21})$$

and

$$\overline{\chi^s} = R^s \overline{T} - \overline{\kappa} \overline{\epsilon^s}. \quad (\text{B.22})$$

Equation (B.21) is replaced by

$$\overline{\kappa} = \frac{\sum_s \overline{\alpha^s} R^s}{\sum_s \overline{\alpha^s} \dot{\epsilon}^s} \quad (\text{B.23})$$

when $\Delta T \rightarrow 0$.

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